Exercise 8.2

(1) (i) Not direction cosines (ii) direction cosines (iii) Not direction cosines

(2) (i) $\left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$ (ii) $\left(\frac{3}{\sqrt{19}}, \frac{-1}{\sqrt{19}}, \frac{3}{\sqrt{19}}\right)$ (iii) (0, 0, 1)

(3) (i) $\left(\frac{3}{\sqrt{89}}, \frac{-4}{\sqrt{89}}, \frac{8}{\sqrt{89}}\right)$, $(3,-4,8)$ (ii) $\left(\frac{3}{\sqrt{11}}, \frac{1}{\sqrt{11}}, \frac{1}{\sqrt{11}}\right)$, $(3,1,1)$

(iii) (0, 1, 0) and (0, 1, 0) (iv) $\left(\frac{5}{\sqrt{2338}}, \frac{-3}{\sqrt{2338}}, \frac{-48}{\sqrt{2338}}\right)$ and $(5,-3-48)$

(v) $\left(\frac{3}{\sqrt{34}}, \frac{4}{\sqrt{34}}, \frac{-3}{\sqrt{34}}\right)$ and $(3,4,-3)$ (vi) $\left(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}\right)$ and $(1,0,-1)$

(4) $\left(\frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right), \left(\frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)$ and $\left(\frac{1}{\sqrt{6}}, \frac{1}{\sqrt{6}}, \frac{-2}{\sqrt{6}}\right)$

(5) $a = \pm\frac{1}{2}$ (6) $a=-1, b=2, c=-1$, or $a=1, b=-2, c=1$ (8) $\lambda = \frac{2}{3}$

(11) (i) $\sqrt{41}, \left(\frac{2}{\sqrt{41}}, \frac{1}{\sqrt{41}}, \frac{-6}{\sqrt{41}}\right)$ (ii) $\sqrt{1123}, \left(\frac{-15}{\sqrt{1123}}, \frac{27}{\sqrt{1123}}, \frac{13}{\sqrt{1123}}\right)$

(12) $\sqrt{44}+\sqrt{218}+\sqrt{110}$ (13) $\frac{1}{\sqrt{398}}(17\hat{i}-3\hat{j}-10\hat{k})$

(14) yes (16) $m = \pm\frac{1}{\sqrt{3}}$

Exercise 8.3

(1) (i) 9 (ii) 4 (2) (i) $\lambda = \frac{5}{2}$ (ii) $\lambda = -2$ (3) $\theta = \frac{\pi}{4}$

(4) (i) $\theta = \cos^{-1}\left(\frac{-9}{49}\right)$ (ii) $\theta = \frac{2\pi}{3}$ (5) $\theta = \frac{2\pi}{3}$ (8) -55

(11) $5\sqrt{2}$ (12) $\frac{41}{7}$ (13) 5 (14) -42

Exercise 8.4

(1) $\sqrt{507}$ (3) $\frac{\pm 10\sqrt{3}}{\sqrt{35}}(5\hat{i}-3\hat{j}+\hat{k})$ (4) $\pm\frac{(-\hat{i}+2\hat{j}-\hat{k})}{\sqrt{6}}$

(5) $8\sqrt{3}$ sq. units (6) $\frac{1}{2}\sqrt{165}$ sq. units (10) $\frac{\pi}{3}$

Exercise 8.5

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
3	3	4	2	2	3	4	4	2	3	2	1	1	1	3
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
4	1	4	4	1	3	2	4	3	2					

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